

JIANGSU LUYE AGROCHEMICALS CO.,LTD

CERTIFICATE OF ANALYSIS

PRODUCT NAME	THIAMETHOXAM 25% WG	
BATCH NO.:	20210310	
QUANTITY:	12000KGS	
MANUFACTURER NAME:	JIANGSU LUYE AGROCHEMICALS CO., LTD.	
PRODUCTION DATE:	MAR.15,2021	
EXPIRY DATE:	MAR.14,2023	
REPORT DATE:	MAR.15,2021	
INSPECTION ITEMS		
ITEMS	INDEX	RESULTS
APPEARANCE	Off-white Granule	Off-white Granule
THIAMETHOXAM, %	≥25.0	25.02
SUSPENSIBILITY, %	≥85.0	97
MOISTURE, %	≤3.0	1.7
WETTING TIME / S	≤90	90
DISINTEGRATION TIME / S	≤60	38
SIEVE ANALYSIS (45μm SIEVE), %	≥98	98.9
RESULT:		

For and on behalf of

SIGNATURE:_____

Jiangsu Luye Agrochemcials Co.,Ltd China

Authorized Signature

JIANGSU LUYE AGROCHEMICALS CO.,LTD

MATERIAL SAFETY DATA SHEET OF THIAMETHOXAM 25% WG

1. IDENTIFICATION OF THE CHEMICAL PRODUCT AND COMPANY

Supplier:JIANGSU LUYE AGROCHEMICALS CO.,LTD

Street address: ECO CHEMICAL INDUSTRIAL PARK, FUNING COUNTY, YANCHENG CITY, JIANGSU PROVINCE, PR CHINA

TEL: +86 21 32568898

Product name: Thiamethoxam 25% WG

Product use: Seed treatment insecticide product for control of certain insect pests in registered crops

2. COMPOSITION/INFORMATION ON INGREDIENTS

Formulation Type: Water dispersible granules

Active Ingredients: Thiamethoxam

Chemical Abstracts name:

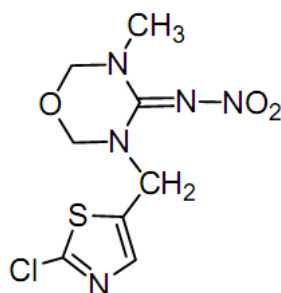
3-[(2-chloro-5-thiazolyl)methyl]tetrahydro-5-methyl-N-nitro-4H-1,3,5-oxadiazin-4-imine

CAS NO. : 153719-23-4

Molecular Formula: C₈H₁₀ClN₅O₃S

Molecular Weight: 291.7

Structural Formula:



Other ingredients determined not to be hazardous

INGRIDIENT	CAS NO	PURITY
thiamethoxam	153719-23-4	≥25%
others	Not available	≤75%

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3. HAZARDS IDENTIFICATION

Symptoms of Acute Exposure: May cause mild eye and skin irritation. Toxic if ingested in large quantity.

Hazardous Decomposition Products: During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

Physical Properties

Appearance: Beige to brown granules.

Odour: Musty.

Unusual Fire, Explosion and Reactivity Hazards: This product is a combustible powder and like all combustible powders can ignite, burn and form explosive mixtures with air if not handled correctly. Mixtures of powder in air with flammable solvent vapours should be avoided. This product has a minimum ignition energy between 30 and 100 millijoules. Mechanical sparks, open

flames and certain hot surfaces can serve as ignition sources for this material. Eliminate the presence of mechanical sparks and other ignition sources where dust clouds of this material could form. This product can energetically decompose at approximately 140 °C (284 °F). Do not store or process at temperatures above 70 °C (158 °F). Do not store near sources of heat including steam lines.

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

4. FIRST AID MEASURES

General: If poisoning is suspected, immediately contact the poison information centre, doctor or nearest hospital. Have the product container, label or Material Safety Data Sheet with you when going for treatment. Tell the person contacted the complete product name, and the type and amount of exposure. Describe any symptoms and follow the advice given.

In case of suspected poisoning: Immediately call a physician.

Eye Contact: Flush eyes with clean water, holding eyelids apart for a minimum of 20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Go to a poison control center or doctor for treatment advice. Obtain medical attention immediately if irritation persists.

Skin Contact: Immediately remove contaminated clothing and wash skin, hair and fingernails thoroughly with soap and water. Flush skin with running water for a minimum of 20 minutes. Obtain medical attention if irritation occurs.

INHALATION: Remove victim to fresh air. If not breathing, give artificial respiration, preferably mouth-to-mouth. If breathing is laboured, give oxygen. Obtain immediate medical attention.

INGESTION: If swallowed, immediately contact poison control doctor or nearest hospital for treatment advice. Provided the patient is conscious, wash out mouth with water. Do not give anything by mouth to an unconscious person. Do not induce vomiting unless directed by a physician or a poison control center. If spontaneous vomiting occurs, have victim lean forward with head down to avoid breathing in of vomitus, rinse mouth and administer water.

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Medical Instructions

Antidote: No specific antidote is known! Apply symptomatic therapy.

5. FIRE FIGHTING MEASURES

Flash point and method: Not applicable.

Upper and lower flammable (explosive) limits in air: Not available.

Auto-ignition temperature: Not available.

Flammability: Not applicable.

Hazardous combustion products: During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

Conditions under which flammability could occur: This product is a combustible powder and like all combustible powders can ignite, burn and form explosive mixtures with air if not handled correctly. Mixtures of powder in air with flammable solvent vapours should be avoided. This product has a minimum ignition energy between 30 and 100 millijoules. Mechanical sparks, open flames and certain hot surfaces can serve as ignition sources for this material. Eliminate the presence of mechanical sparks and other ignition sources where dust clouds of this material could form. This product can energetically decompose at approximately 140 °C (284 °F). Do not store or process at temperatures above 70 °C (158 °F). Do not store near sources of heat including steam lines. Keep fire exposed containers cool by spraying with water.

Extinguishing media: Use foam, carbon dioxide, dry powder, halon extinguishant or water fog or mist, (avoid use of water jet). Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. Contain run-off water with, for example, temporary earth barriers.

Sensitivity to explosion by mechanical impact: Low.

Sensitivity to explosion by static discharge: Low.

6. ACCIDENT RELEASE MEASURES

Personal Precautions: Make sure all personnel involved in the spill cleanup follow good industrial hygiene practices. A small spill can be handled routinely. Wear suitable protective clothing and eye protection to prevent skin and eye contact. Use adequate ventilation and wear an air-supplied respirator to prevent inhalation.

Procedures for dealing with release or spill: Control the spill at its source. Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of water. Clean up spills immediately, observing precautions outlined in Sections 7 and 8. Scoop or sweep up material and place into a disposal container. Wash area with detergent and water. Pick up wash liquid with additional absorbent and place

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into compatible disposal container. On soils, skim off the upper contaminated layer and collect for disposal. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition. Spillages or uncontrolled discharges into watercourses must be alerted to the appropriate regulatory body.

7. HANDLING AND STORAGE

Handling practices: This product is a combustible powder and like all combustible powders can ignite, burn and form explosive mixtures with air if not handled correctly. Mixtures of powder in air with flammable solvent vapours should be avoided. This product has a minimum ignition energy between 30 and 100 millijoules. Mechanical sparks, open flames and certain hot surfaces can serve as ignition sources for this material. Eliminate the presence of mechanical sparks and other ignition sources where dust clouds of this material could form. This product can energetically decompose at approximately 140 °C (284 °F). Do not store or process at temperatures above 70 °C (158 °F). Do not store near sources of heat including steam lines. Keep fire exposed containers cool by spraying with water.

Keep out of reach of children and animals. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. After work, rinse gloves and remove protective equipment. Wash hands thoroughly with soap and water after handling, and before eating, tobacco use, drinking, or using the toilet. Wash contaminated clothing before re-use and separate from household laundry. Keep containers closed when not in use. Keep product, wash or rinse water, and contaminated materials out of water, away from crops, and away from access by people, animals and birds.

Appropriate storage practices/requirements: Store in original container only in a well-ventilated, cool, dry, secure area.

Protect from heat, sparks and flame. Do not expose sealed containers to temperatures above 40 °C and prevent product from freezing. Keep separate from other products to prevent cross contamination. Rotate stock. Clean up spilled material immediately.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingestion: Do not eat, drink, handle tobacco, or apply cosmetics in areas where there is a potential for exposure to this material. Always wash thoroughly after handling.

Eye Contact: Where eye contact is likely, use chemical splash goggles. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

Skin Contact: Where contact is likely, wear chemical-resistant gloves (such as nitrile or butyl), coveralls, socks and chemical-resistant footwear. For overhead exposure, wear chemical-resistant headgear.

Inhalation: A respirator is not normally required when handling this substance. Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below exposure limits. Protection provided

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by air-purifying respirators is limited. Use a pressure demand atmosphere-supplying respirator if there is any potential for uncontrolled release, exposure levels are not known, or under any other circumstances where air-purifying respirators may not provide adequate protection.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Beige to brown granules.

Formulation Type: Wettable granule.

Odour: Musty.

pH: 9.4 (1% aqueous solution).

Vapour pressure and reference temperature: 2×10^{-11} mmHg @ 20 °C (Thiamethoxam Technical).

Boiling point: Not available.

Melting point: Not available.

Freezing point: Not available.

Solubility in Water: 4.1 g/L @ 25 °C (Thiamethoxam Technical).

10. STABILITY AND REACTIVITY

Stability: Stable under normal use and storage conditions.

Conditions to Avoid: None known.

Hazardous Decomposition Products: Can decompose at high temperatures forming toxic gases.

Hazardous Polymerization: Will not occur.

11. TOXICOLOGICAL INFORMATION

Ingestion: Practically Non-Toxic Oral (LD50 Rat): > 5,000 mg/kg body weight

Dermal: Slightly Toxic Dermal (LD50 Rat): > 2,000 mg/kg body weight

Inhalation: Practically Non-Toxic Inhalation (LC50 Rat): > 2.79 mg/L air - 4 hours

Eye Contact: Mildly Irritating (Rabbit)

Skin Contact: Slightly Irritating (Rabbit)

Skin Sensitization: Not sensitizing (guinea pig)

12. ECOLOGICAL INFORMATION

The active ingredient, thiamethoxam, is slightly to practically non-toxic to fish, birds and aquatic invertebrates (water flea) but is highly toxic to the honeybee and one species of aquatic invertebrate (chironomid). However, exposure and risk to non-target species to thiamethoxam is minimal and acceptable based on the low application rates and application methodologies.

Eco-Acute Toxicity (Thiamethoxam Technical):

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Bees LC ₅₀ /EC ₅₀	0.024 µg/bee
Invertebrates (<i>Daphnia magna</i>) 48-hour LC ₅₀ /EC ₅₀	>100 ppm
Invertebrates (Chironomid) 48-hour LC ₅₀ /EC ₅₀	0.035 ppm
Fish (Rainbow Trout) 96-hour LC ₅₀ /EC ₅₀	> 100 ppm
Fish (Bluegill) 96-hour LC ₅₀ /EC ₅₀	> 114 ppm
Birds (8-day dietary - Bobwhite Quail) LC ₅₀ /EC ₅₀	> 5,200 ppm
Birds (8-day dietary - Mallard Duck) LC ₅₀ /EC ₅₀	> 5,200 ppm
Bobwhite Oral LD ₅₀	1,552 mg/kg
Mallard Oral LD ₅₀	576 mg/kg

Eco-Chronic Toxicity (Thiamethoxam Technical):

Fish (Fathead minnow) Early Life NOEC	> 20 mg/L
Invertebrate (<i>Daphnia magna</i>) Life Cycle NOEC	> 100 mg/L
Invertebrate (Chironomid) Life Cycle NOEC	0.010 mg/L
Mallard Reproduction NOEC	300 ppm
Bobwhite Reproduction NOEC	900 ppm

Environmental Fate

The active ingredient, thiamethoxam, has a low bioaccumulation potential, low mobility, and moderate persistence in soil and water. The dissipation half-life in soil is 48-239 days. The main route of degradation is by microbial degradation and formation of bound residues.

13. DISPOSAL CONSIDERATION

Do not reuse empty containers. Empty container retains product residue. Triple rinse, or equivalent, empty container, return rinse water to dilution mixture, and dispose of dilution mixture as a hazardous waste if it cannot be disposed of by use according to label instructions. Dispose of empty containers in accordance with local regulations. Consult provincial environment ministry for advice on waste disposal. Industrial/commercial waste may be handled at licensed facilities only. Waste shipments must be securely packaged and properly labelled. Only licensed carriers may be used, and proper documents must accompany the shipment.

14. TRANSPORT INFORMATION

UN Number: 3077.

UN Proper shipping name: Environmentally hazardous substance, solid, n.o.s. (contain thiamethoxam)

Transport hazard class: Class 9

Packing group: III

Marine pollutant: Yes.

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15. REGULATORY INFORMATION

Keep locked up out of reach of children and other, unauthorized persons. Keep away from food, drink and animal feeding stuffs.

Users should wash hands before eating, drinking, chewing gum, using tobacco or using the toilet. Do not get in eyes, on skin or on clothing. Do not breathe dust, vapor or spray mist.

In case of accident or if you feel unwell, seek medical advice immediately.

This material and its container must be disposed of in a safe way.

16. OTHER INFORMATION

This MSDS summarizes our best knowledge of the health and safety hazard information of the product and how to safely handle and use the product in the workplace. Each user should read this MSDS and consider the information in the context of the how the product will be handled and used in the workplace including in conjunction with other products.

If clarification or further information is needed to ensure that an appropriate risk assessment can be made the user should contact this company.

END OF MSDS

For and on behalf of

Jiangsu Luye Agrochemicals Co.,Ltd China

Amy

Authorized Signature